👩‍🏫 Staff Guide · Biology A · Algae & Photosynthesis (Level 3)

Level 3 = resource-room students who can work close to the gen ed lesson. Same topics and order as Ms. Dunham's class; more reading and writing than Level 2.

1. How the pieces line up

Class lesson	Level 3 piece
A1 A look at algae (microscope)	Reading part 1; notes section A (magnification math)
A4 Pop bead photosynthesis + diagram	Reading parts 2-3; notes section B
A4.5 Stomata peel; A4 Elodea & snails; A2 Algae growth lab	Reading parts 4-5; notes section C
Photosynthesis video quiz / writing practice	Quick check + unit game

2. Supports built in

- Reading and notes each have a web copy with a 🔊 Read aloud button (highlights as it reads)
- Key-word box at the top of the notes; word problems have a worked example first
- Less filled in than Level 2: short answers, tables, a drawing
- No slide decks: students follow Ms. Dunham's slides with the Level 3 notes

3. Files

- Reading (+ 🔊 web copy)
- Guided notes + key (+ 🔊 web copy)
- Quick check + key
- Unit game: 10 questions

4. What to grade

- Guided notes stand in for the matching class notes or worksheet (same points when complete)
- Quick check: score it, then scale to the class quiz points
- Reading check questions: completion points
- Modified grading only as the IEP allows. Check with the case manager.

5. Who to ask

- Mr. Driver (SLE): changes and IEP questions.
- Class content: Ms. Dunham.

🌿 Algae & Photosynthesis: Guided Notes

Biology A · Unit 2 · Level 3 (closer to the class work) · goes with Unit 2 Algae Biofuel: A1 algae + microscope, A2 algae lab, A4 Pop beads, Elodea & snails, A4.5 Stomata

Name: _____

Use the class slides and your reading. Key words:

algae · *photosynthesis* · *glucose* · *chloroplast* · *chlorophyll* · *stomata* · *guard cells* · *BTB* · *magnification*

A. Microscopes (A1)

1. **Total magnification = eyepiece × objective**

> Example: 10x eyepiece × 10x objective = 100x

a) 10x eyepiece × 4x objective = _____

b) 10x eyepiece × 40x objective = _____

2. Always find your object on _____ power first.

B. Photosynthesis (A4 Pop beads)

3. Photosynthesis happens in the _____.

4. Light energy becomes stored _____ energy in glucose.

5. Fill in the table.

Inputs (go in)	Outputs (come out)
	—

6. What happens in the light reactions?

C. Stomata and BTB (A4.5, Elodea & snails, A2)

7. Guard _____ open and close the stomata.

8. A tube has a snail and BTB in the dark. Predict the color after a day and explain.

9. In the algae lab, which tube grew the most algae and why?

🌿 Algae & Photosynthesis: Reading

Biology A · Unit 2 · Level 3 (closer to the class work) · goes with Unit 2 Algae Biofuel: A1 algae + microscope, A2 algae lab, A4 Pop beads, Elodea & snails, A4.5 Stomata

Name: _____

How Algae Turn Light into Food

Algae are plant-like living things that live in water. Some are big, like kelp; most are tiny, like phytoplankton and pond scum. To see tiny algae, we use a microscope. Total magnification is the eyepiece (10x) times the objective lens. A 40x objective gives 40 × 10 = 400x. Always find the object on low power first, and clean lenses only with lens paper.

Algae and plants make their own food through photosynthesis. They take in carbon dioxide and water and use light energy to make glucose, a sugar that stores chemical energy. Oxygen is released. The word equation is: carbon dioxide + water + light → glucose + oxygen.

Photosynthesis happens in the chloroplasts, which contain green chlorophyll. It has two stages. In the light reactions (in the thylakoids), light energy splits water and oxygen is released. In the Calvin cycle (in the stroma), carbon dioxide is built into glucose.

Leaves have tiny pores called stomata, mostly on the underside. Guard cells open and close the stomata. Carbon dioxide comes in, and oxygen and water vapor go out. In hot, dry weather the stomata close to save water.

BTB is a color indicator for carbon dioxide. It turns yellow when CO2 goes up and blue when CO2 goes down. Snails make CO2 by respiration; Elodea in the light uses CO2 by photosynthesis. In the algae lab, fertilizer helped algae grow, but dish soap did not.

Check your reading

1. Write the word equation for photosynthesis.

2. What do guard cells do?

3. A tube of BTB turns yellow. What does that tell you?

🌿 Algae & Photosynthesis: Quick Check

Biology A · Unit 2 · Level 3 (closer to the class work) · goes with Unit 2 Algae Biofuel: A1 algae + microscope, A2 algae lab, A4 Pop beads, Elodea & snails, A4.5 Stomata

Name: _____

Multiple choice and short answer. You may have it read aloud.

1. Which is an OUTPUT of photosynthesis?

A. Oxygen | B. Carbon dioxide | C. Light

2. Photosynthesis happens in the…

A. Mitochondria | B. Stomata | C. Chloroplasts

3. The 40x objective with a 10x eyepiece gives…

A. 50x | B. 40x | C. 400x

4. BTB turns yellow when…

A. CO2 goes down | B. CO2 goes up | C. Light goes up

5. Why do stomata close on hot, dry days?

6. Explain why algae are at the base of the food web.

🌿 Algae & Photosynthesis: Guided Notes · ANSWER KEY

Biology A · Unit 2 · Level 3 (closer to the class work) · goes with Unit 2 Algae Biofuel: A1 algae + microscope, A2 algae lab, A4 Pop beads, Elodea & snails, A4.5 Stomata

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algae | *photosynthesis* | *glucose* | *chloroplast* | *chlorophyll* | *stomata* | *guard cells* | *BTB* | *magnification*

A. Microscopes (A1)

1. **Total magnification = eyepiece × objective**

> Example: 10x eyepiece × 10x objective = 100x

a) 10x eyepiece × 4x objective = **40x**

b) 10x eyepiece × 40x objective = **400x**

2. Always find your object on **low** power first.

B. Photosynthesis (A4 Pop beads)

3. Photosynthesis happens in the **chloroplasts**.

4. Light energy becomes stored **chemical** energy in glucose.

5. Fill in the table.

Inputs (go in)	Outputs (come out)
carbon dioxide	**glucose**
water	**oxygen**
light energy	—

6. What happens in the light reactions?

Light energy splits water and oxygen is released (in the thylakoids).

C. Stomata and BTB (A4.5, Elodea & snails, A2)

7. Guard **cells** open and close the stomata.

8. A tube has a snail and BTB in the dark. Predict the color after a day and explain.

Yellow: the snail does respiration and adds CO2; no light means no photosynthesis to use it.

9. In the algae lab, which tube grew the most algae and why?

The fertilizer tube; fertilizer gives nutrients that help algae grow.

🔑 *Key written by our team from Ms. Dunham's materials, please check.*

🌿 Algae & Photosynthesis: Reading · ANSWER KEY

Biology A · Unit 2 · Level 3 (closer to the class work) · goes with Unit 2 Algae Biofuel: A1 algae + microscope, A2 algae lab, A4 Pop beads, Elodea & snails, A4.5 Stomata

Name: _____

How Algae Turn Light into Food

Algae are plant-like living things that live in water. Some are big, like kelp; most are tiny, like phytoplankton and pond scum. To see tiny algae, we use a microscope. Total magnification is the eyepiece (10x) times the objective lens. A 40x objective gives 40 × 10 = 400x. Always find the object on low power first, and clean lenses only with lens paper.

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BTB is a color indicator for carbon dioxide. It turns yellow when CO2 goes up and blue when CO2 goes down. Snails make CO2 by respiration; Elodea in the light uses CO2 by photosynthesis. In the algae lab, fertilizer helped algae grow, but dish soap did not.

Check your reading

1. Write the word equation for photosynthesis.
Carbon dioxide + water + light → glucose + oxygen.

2. What do guard cells do?
They open and close the stomata.

3. A tube of BTB turns yellow. What does that tell you?
Carbon dioxide went up.

🌿 Algae & Photosynthesis: Quick Check · ANSWER KEY

Biology A · Unit 2 · Level 3 (closer to the class work) · goes with Unit 2 Algae Biofuel: A1 algae + microscope, A2 algae lab, A4 Pop beads, Elodea & snails, A4.5 Stomata

Name: _____

Multiple choice and short answer. You may have it read aloud.

1. Which is an OUTPUT of photosynthesis?

A. Oxygen | B. Carbon dioxide | C. Light

2. Photosynthesis happens in the…

A. Mitochondria | B. Stomata | **C. Chloroplasts**

3. The 40x objective with a 10x eyepiece gives…

A. 50x | B. 40x | **C. 400x**

4. BTB turns yellow when…

A. CO2 goes down | **B. CO2 goes up** | C. Light goes up

5. Why do stomata close on hot, dry days?

To save water (less water vapor escapes).

6. Explain why algae are at the base of the food web.

They are producers: they make glucose from light by photosynthesis.

🔑 *Key written by our team from Ms. Dunham's quizzes and notes, please check.*